Macroeconomics #6

Dynamics and Modelling

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Today's lecture

- Last lecture, we studied OLG
- We have only touched a case of monotonic conversion of one state variable, but it is (mostly) boring
- Today, we study some richer dynamics of one state variable, built on the OLG framework

Agenda

1. Two-period: New Tech
2. Two-period: Debt
3. OLG: Human Capital
4. OLG: Occupational Choice
5. OLG: Subsistence consumption
6. OLG: Impatience Trap
7. OLG: Positive Externality

Investment in New Tech

Model

- A two-period closed economy can invest in physical capital k_1 and new technology k_e.
- New technology requires capital: λk_e (with $\lambda > 1$, relative price).
- If adopted, period-1 TFP becomes $A(k_e)$ with $A' > 0$, A concave, $A(0) = 1$.

Production: $y_0 = f(k_0)$, $y_1 = A(k_e)\, f(k_1)$.

$$\text{s.t.} \quad c_0 + k_1 + \lambda k_e = f(k_0), \qquad c_1 = A(k_e)\, f(k_1).$$

Optimization

The country solves:

$$\max_{k_1, k_e} \{u(f(k_0) - k_1 - \lambda k_e) + \beta u(A(k_e)\, f(k_1))\}.$$

FOCs:

$$u'(c_0) = \beta u'(c_1)\, A(k_e^*)\, f'(k_1^*) \qquad \text{(w.r.t. } k_1\text{)} \tag{1}$$

$$\lambda\, u'(c_0) = \beta u'(c_1)\, A'(k_e^*)\, f(k_1^*) \qquad \text{(w.r.t. } k_e\text{)} \tag{2}$$

Dividing: $\lambda \dfrac{f'(k_1^*)}{f(k_1^*)} = \dfrac{A'(k_e^*)}{A(k_e^*)}$. This equates the relative marginal return of the two capital types.

Equilibrium & Dynamics

Example: $u = \ln(c)$, $f(k) = \gamma k$, $A(k) = ak + 1$:

$$k_1^* = \frac{\beta}{1+2\beta}\left(\gamma k_0 + \frac{\lambda}{a}\right), \qquad k_e^* = \frac{1}{1+2\beta}\left(\frac{\beta\gamma}{\lambda}k_0 - \frac{1+\beta}{a}\right).$$

Condition for $k_e^* > 0$:

$$a\beta\gamma k_0 > \lambda(1+\beta).$$

- The country invests in new technology iff it is rich enough (k_0 high) or the quality of technology is good (a high).
- This is a threshold effect: below a critical initial capital, the country cannot afford the fixed (relative) cost of technology adoption.

Debt

Model

Setup: Open developing country, two periods.

- Borrows E in period -1, serving as additional capital.
- Reimbursement: $(1+r)E$ at end of period 0.
- Production: $F(k) = Ak$ (linear), depreciation δ.

Constraints:

$$c_0 + k_1 - (1-\delta)(k_0 + E) = A(k_0 + E) - (1+r)E,$$

$$c_1 = A(k_1) + (1-\delta)k_1.$$

Optimization

Country's problem:

$$\max_{k_1} \ \{u(c_0) + \beta u(c_1)\},$$

with $u(c) = c^\theta/\theta$.
FOC:

$$u'(c_0) = \beta u'(c_1)(A + 1 - \delta).$$

Optimal capital:

$$\tilde{k}_1 = \frac{\beta^{\frac{1}{1-\theta}} \left[(A + 1 - \delta)k_0 + (A - r - \delta)E\right](A + 1 - \delta)^{\frac{\theta}{1-\theta}}}{1 + \beta^{\frac{1}{1-\theta}} (A + 1 - \delta)^{\frac{\theta}{1-\theta}}}.$$

Condition for debt to enhance growth:

$$\frac{\partial \tilde{k}_1}{\partial E} = \underbrace{\frac{C}{1+C}}_{>0}(A - r - \delta)$$

where $C = [\beta(1 + A - \delta)^{\theta}]^{1/(1-\theta)} > 0$. Borrowing helps growth iff:

$$\boxed{A > r + \delta}.$$

Interpretation: The marginal productivity of capital must exceed the total cost of borrowing (interest r + depreciation δ).

Poverty-trap implication (Azariadis 4.2): In an economy with external IRS, even a small amount of public debt can crowd out private capital enough to move the economy from a perpetual-growth path onto a stagnation path. Public debt displaces claims on private capital from household portfolios.

Human Capital

Model

- **Setup**: Two-period OLG
- Old parents allocate income between consumption c_t and child's education e_t.
- Single final good produced linearly from human capital: $Y_t = H_t$.
- Wage per efficiency unit normalized to 1.
- **Key insight**: The poor require relatively higher returns to increase investment in education (γ parameter below), because the marginal utility of consumption is high.

Human-capital technology:

$$h_{t+1} = f(e_t) = \log(\gamma e_t + \nu), \qquad \gamma > 0,\ \nu > 1.$$

- $\mu \equiv \log \nu$ = human capital with zero education.
- Bounded returns at zero: $f'(0) = \gamma/\nu < \infty$.
- Log-form ensures diminishing returns everywhere.

Preferences:

$$u_t = \log c_t + \delta\, h_{t+1}, \qquad \delta > 0.$$

- MRS between consumption and child's education: decreases with parental h_t.
- Poor parents require higher marginal returns to start investing in education.

Optimization

Parents solve:

$$\max_{e_t \geq 0} \left\{ \log(h_t - e_t) + \delta \log(\gamma e_t + \nu) \right\}.$$

Lagrangian (with non-negativity):

$$\mathcal{L} = \log(h_t - e_t) + \delta \log(\gamma e_t + \nu) + \lambda e_t, \qquad \lambda \geq 0, \ \lambda e_t = 0.$$

Interior FOC ($\lambda = 0$):

$$\frac{\partial \mathcal{L}}{\partial e_t} = -\frac{1}{h_t - e_t} + \frac{\delta \gamma}{\gamma e_t + \nu} = 0.$$

Solving:

$$e_t = \frac{\delta}{1+\delta} h_t - \frac{\nu}{\gamma(1+\delta)}.$$

Corner solution ($e_t = 0$) when:

$$h_t \leq \underline{h} \equiv \frac{\nu}{\delta \gamma}.$$

Dynamics

The kinked education function:

$$e_t^*(h_t) = \begin{cases} 0 & h_t \leq \underline{h}, \\ \dfrac{\delta}{1+\delta} h_t - \dfrac{\nu}{\gamma(1+\delta)} & h_t > \underline{h}. \end{cases}$$

Substituting e_t^* into the human-capital technology:

$$h_{t+1} = \phi(h_t) = \begin{cases} \mu & h_t \leq \underline{h}, \\ \log \dfrac{\delta(\gamma h_t + \nu)}{1+\delta} & h_t > \underline{h}. \end{cases}$$

Conditions for multiple steady states

1. Condition 1 (slope at threshold): $\phi'(\underline{h}^+) > 1$. (why?)
2. Condition 2 (lowest steady state): $\phi(h_t < \underline{h}) < \underline{h}$. (why?)
3. Condition 3 (upper branch reaches the 45° line): $\max_{h>\underline{h}}[\phi(h) - h] > 0$.

Conditions 1–2 give $\frac{\nu}{\delta}\frac{1}{\ln(\nu)} > \gamma > \frac{\nu}{\delta}(1+\delta)$; Condition 3 (the maximum is at $h = 1 - \nu/\gamma$) gives $\ln\frac{\delta\gamma}{1+\delta} + \frac{\nu}{\gamma} > 1$.

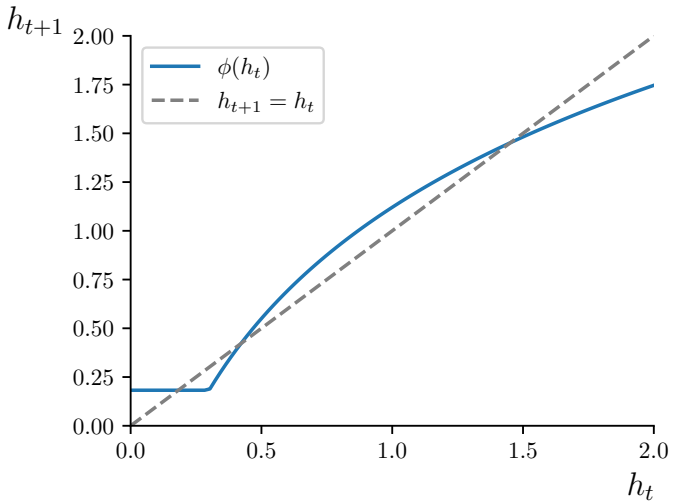


Figure:

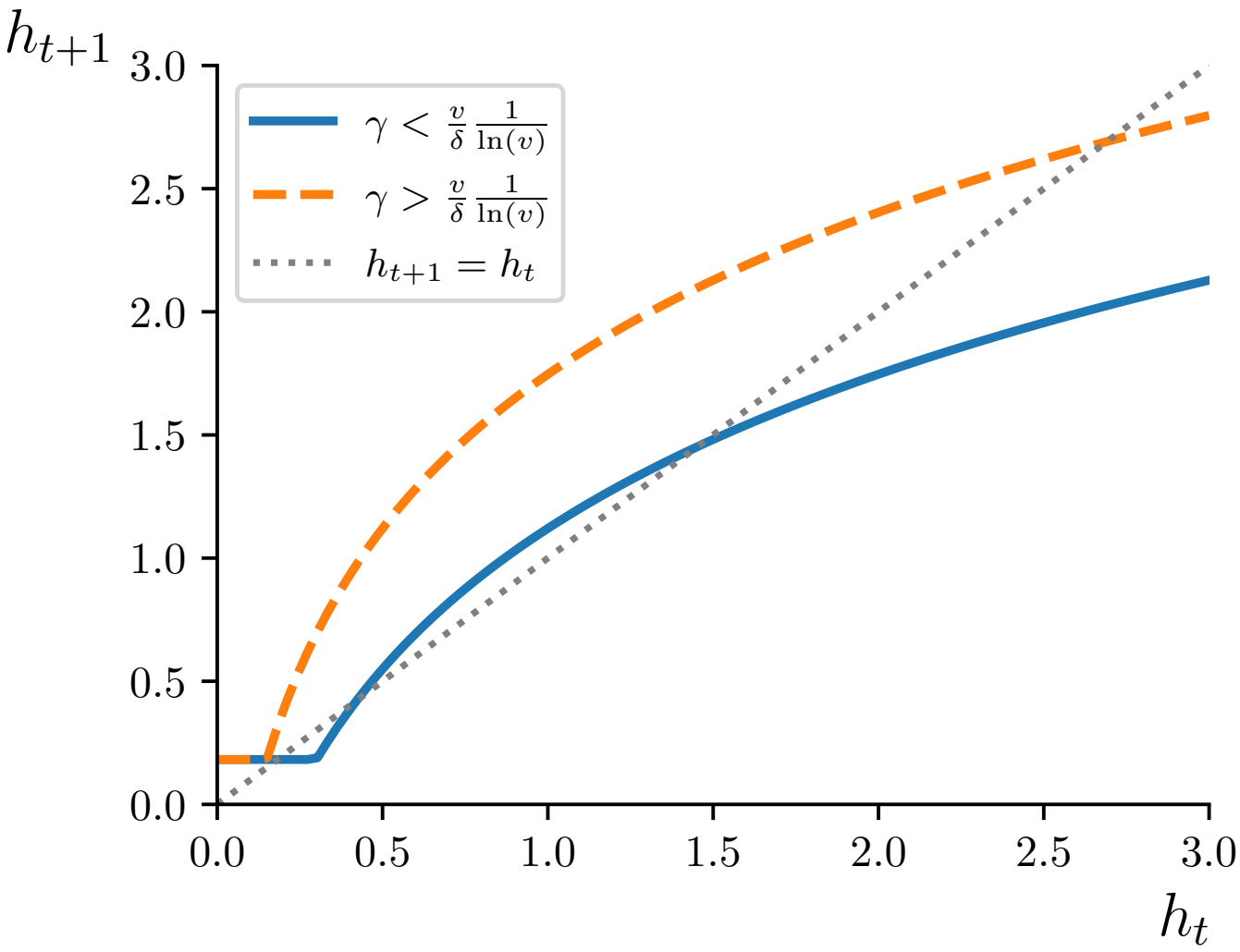


Figure: When γ is too big

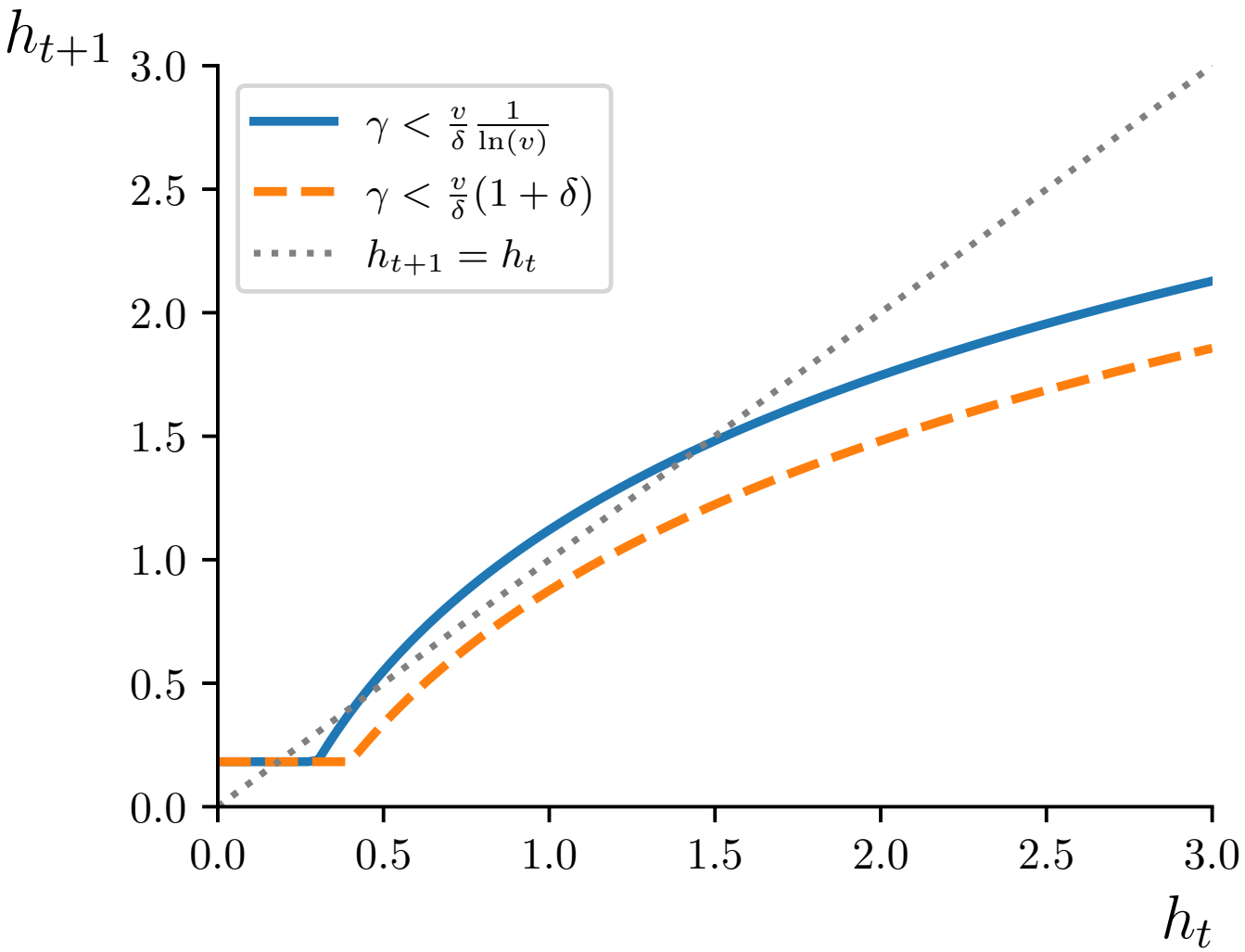


Figure: When γ is too small

Occupational Choice

Model Construction

- Three-period OLG: childhood, adulthood, old age.
- In adulthood: allocate 1 unit of time to (i) work, (ii) child rearing, (iii) optional schooling.
- Three factors: K, skilled labour L^s, unskilled labour L^u.
- **Capital–skill complementarity**: K and L^s are complements; L^u is a perfect substitute for the K–L^s aggregate.
- Agents choose whether to become skilled (cost τ of time) or remain unskilled, and how many children to have (cost z per child).

Technology

Cobb–Douglas–linear hybrid production function:

$$Y_t = A\big[K_t^{\alpha}(L_t^s)^{1-\alpha} + bL_t^u\big], \qquad A > 0,\ b > 0,\ \alpha \in (0,1).$$

Aggregate factor supplies (φ_t = share of skilled):

$$L_t^s = (1 - \tau - zn_t^s)\varphi_t N_t, \qquad L_t^u = (1 - zn_t^u)(1 - \varphi_t)N_t.$$

Competitive wages:

$$w_t^s = A(1-\alpha)\left(\frac{k_t}{(1-\tau-zn_t^s)\varphi_t}\right)^{\alpha}, \qquad w_t^u = Ab.$$

Unskilled wage is independent of the skill composition — the source of the compositional non-convexity.

Households

Preferences (logarithmic over fertility and old-age consumption):

$$u_t^j = \gamma \ln n_t^j + (1-\gamma) \ln c_{t+1}^j, \qquad j \in \{s, u\},\ \gamma \in (0,1).$$

Budget constraints:

$$s_t^s = w_t^s(1 - \tau - z n_t^s), \tag{3}$$
$$s_t^u = w_t^u(1 - z n_t^u). \tag{4}$$

consumption in second period of life

$$c_{t+1}^s = (1 + r_{t+1})\, w_t^s(1 - \tau - z n_t^s), \tag{5}$$
$$c_{t+1}^u = (1 + r_{t+1})\, w_t^u(1 - z n_t^u). \tag{6}$$

Each agent chooses n_t^j and c_{t+1}^j to maximize lifetime utility.

Optimization

Skilled worker solves:

$$\max_{n_t^s} \ \gamma \ln n_t^s + (1-\gamma) \ln \left[(1+r_{t+1}) w_t^s (1-\tau - z n_t^s)\right].$$

FOC:

$$\frac{\gamma}{n_t^s} = \frac{(1-\gamma)z}{1-\tau - z n_t^s} \quad \Longrightarrow \quad n_t^s = \frac{\gamma(1-\tau)}{z}.$$

Similarly for unskilled: $n_t^u = \dfrac{\gamma}{z}$.

Result: $n_t^s < n_t^u$ — skilled workers have strictly fewer children.

- Savings: $s_t^s = w_t^s(1-\tau)(1-\gamma)$, $s_t^u = w_t^u(1-\gamma)$.
- Skilled agents earn higher wages and have fewer children $\Rightarrow$ higher capital per capita.

Equilibrium — Occupational Arbitrage

In equilibrium with both occupations, agents must be indifferent. Equating indirect utilities:

$$\gamma \ln n_t^s + (1-\gamma) \ln c_{t+1}^s = \gamma \ln n_t^u + (1-\gamma) \ln c_{t+1}^u.$$

Simplifying yields the **arbitrage condition**:

$$\frac{w_t^u}{w_t^s} = (1-\tau)^{\frac{1}{1-\gamma}}.$$

Combining with the wage equations gives the skilled share:

$$\varphi(k_t) = \min\{\theta k_t, 1\}, \qquad \theta = \frac{(1-\tau)^{\frac{1}{\alpha(1-\gamma)}-1}}{1-\gamma} \left(\frac{1-\alpha}{b}\right)^{1/\alpha}.$$

As k_t rises $\Rightarrow$ φ rises: capital deepening draws workers into the skilled sector.

Capital Accumulation

Aggregate capital next period equals total savings today:

$$K_{t+1} = N_t\left[\varphi_t s_t^s + (1-\varphi_t)s_t^u\right],$$

with savings $s_t^s = w_t^s(1-\tau)(1-\gamma)$, $s_t^u = Ab(1-\gamma)$, and $w_t^s = Ab/(1-\tau)^{\frac{1}{1-\gamma}}$ from the arbitrage condition.
Population evolves as $N_{t+1} = m_t N_t$, with
$m_t = \varphi_t n_t^s + (1-\varphi_t)n_t^u = \frac{\gamma}{z}(1-\tau\varphi_t)$.

Dynamics — Explicit Law of Motion

$$k_{t+1} = \frac{\varphi_t s_t^s + (1-\varphi_t) s_t^u}{m_t} = \phi(k_t)$$

Below the critical capital ($k_t < \bar{k}$, mixed occupations, $\varphi_t = \theta k_t$):

$$k_{t+1} = \underbrace{\frac{Az(1-\gamma)}{\gamma} \frac{1}{1-\tau\theta k_t} \left[\frac{(1-\alpha)(1-\tau)^{1-\alpha}\theta^{1-\alpha} k_t}{(1-\gamma)^{\alpha}} + b(1-\theta k_t) \right]}_{\phi_{\text{convex}}(k_t)},$$

with $\phi_{\text{convex}}(0) = \frac{Az(1-\gamma)}{\gamma} b > 0$, $\phi''_{\text{convex}} > 0$ on $[0, \bar{k})$.

Above the critical capital ($k_t \geq \bar{k}$, all skilled, $\varphi_t = 1$):

$$k_{t+1} = \underbrace{\frac{Az(1-\gamma)}{\gamma} \frac{(1-\alpha)}{(1-\tau)^{\alpha}(1-\gamma)^{\alpha}} k_t^{\alpha}}_{\phi_{\text{concave}}(k_t)},$$

with $\phi''_{\text{concave}} < 0$ on $[\bar{k}, \infty)$.

Dynamics — Why the Convexity?

The map $\phi(k_t)$ switches from convex to concave at $\bar{k}$.
For $k_t < \bar{k}$ (compositional convexity): As k_t rises, $\varphi_t = \theta k_t$ rises $\Rightarrow$ more workers switch from unskilled to skilled.

- Skilled earn more ($w_t^s > w_t^u$) $\Rightarrow$ higher savings per adult.
- Skilled have fewer children ($n_t^s < n_t^u$) $\Rightarrow$ lower fertility m_t.
- Both forces raise k_{t+1} through numerator $\uparrow$ and denominator $\downarrow$.
- This accelerating effect makes k_{t+1} increase faster than linearly in k_t.

For $k_t \geq \bar{k}$ (Diamond concavity): All workers are skilled, $\varphi_t = 1$, the model collapses to the standard log–log Diamond OLG: $k_{t+1} \propto k_t^{\alpha}$, which is strictly concave.

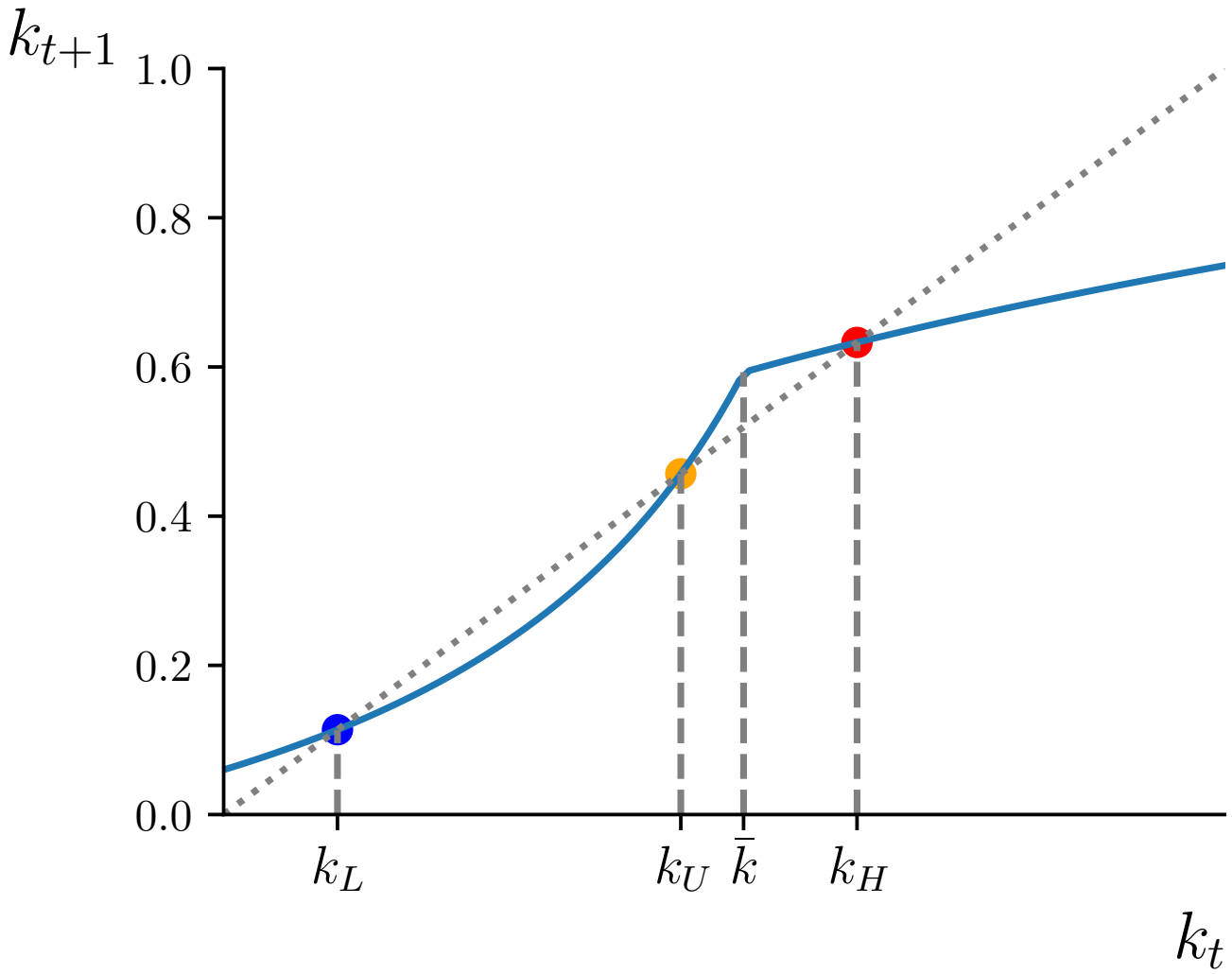


Figure: Dynamics with 3 equilibria

Condition for Three Equilibria

When does the model yield $k_L^* < k_U < \bar{k} < k_H^*$?
Lower bound: $\phi(\bar{k}) > \bar{k}$ — the concave branch lies above the 45° line at $\bar{k}$, so k_{t+1} crosses back down later.

$$A > A_{\min} \equiv \frac{\gamma}{z(1-\gamma)} \cdot \frac{(1-\tau)^\alpha (1-\gamma)^\alpha}{1-\alpha} \cdot \bar{k}^{1-\alpha}$$

Upper bound: $\exists\, k < \bar{k}$ with $\phi(k) < k$ — the convex branch must dip below the 45° line, otherwise no low trap exists.

$$A < A_{\max}$$

where $A_{\max}$ is defined implicitly by $\phi(k; A_{\max}) = k$ and $\phi'(k; A_{\max}) = 1$ for some $k \in (0, \bar{k})$.
Three equilibria: $A_{\min} < A < A_{\max}$.
E.g., $\alpha = 0.33$, $\tau = 0.6$, $\gamma = 0.6$, $z = 0.2$, $b = 0.1 \Rightarrow 3.9 < A < 5.5$. At $A = 4.5$: three equilibria.

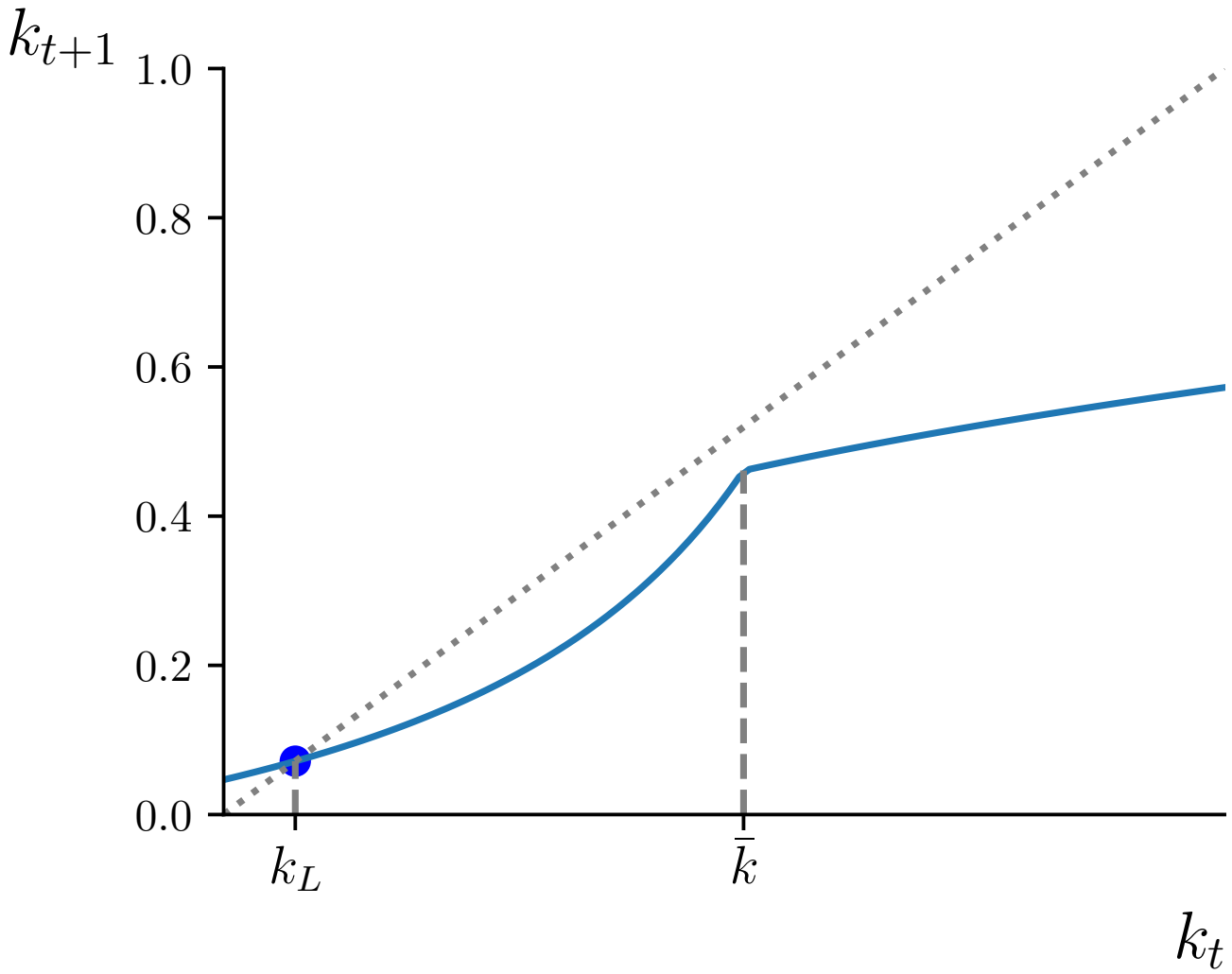


Figure: A is too small

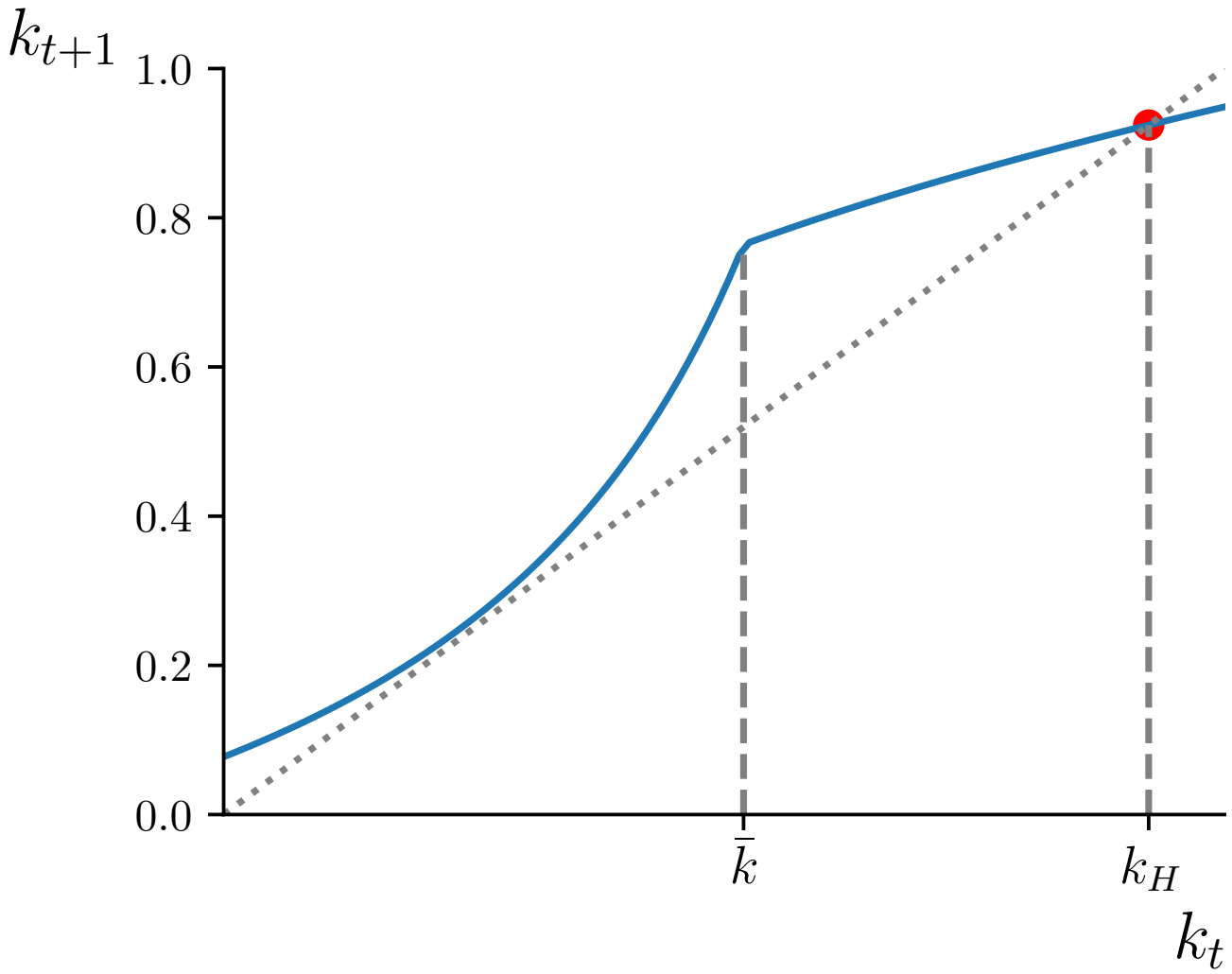


Figure: A is too big

Subsistence

Model Construction

- Two-period OLG with Stone–Geary (subsistence) preferences.
- Constant fertility n, Cobb–Douglas technology $f(k) = k^\alpha$, full depreciation $\delta = 1$.
- **Key ingredient**: old-age subsistence consumption proportional to output per worker.

Preferences:

$$v_t = \log(c_t^1) + \beta \log(c_t^2 - x_t k_{t+1}^\alpha), \qquad c_t^2 \geq x_t k_{t+1}^\alpha,$$

with $x_t = 0$ if $k_t < \hat{k}$, and $x_t = \bar{c}$ if $k_t \geq \hat{k}$.

Optimization

Household problem:

$$\max \ \log c^1 + \beta \log(c^2 - x_t k_{t+1}^{\alpha}), \qquad c^1 + \frac{c^2}{R_{t+1}} = w_t.$$

FOC w.r.t. c^2 (via Lagrangian):

$$\frac{1}{c^1} = \lambda, \qquad \frac{\beta}{c^2 - x_t k_{t+1}^{\alpha}} = \frac{\lambda}{R_{t+1}}.$$

Saving $= w_t - c^1 = sw_t + \frac{x_t k_{t+1}^{\alpha}}{(1+\beta)R_{t+1}}$, and with $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$ this is $sw_t + \frac{x_t k_{t+1}}{\alpha(1+\beta)}$.
With market clearing $nk_{t+1} = s_t$, the law of motion becomes:

$$k_{t+1} = \frac{s(1-\alpha)}{n - (1-s)x_t/\alpha} \, k_t^{\alpha}, \qquad s = \frac{\beta}{1+\beta}.$$

Dynamics and Trap

Piecewise law of motion:

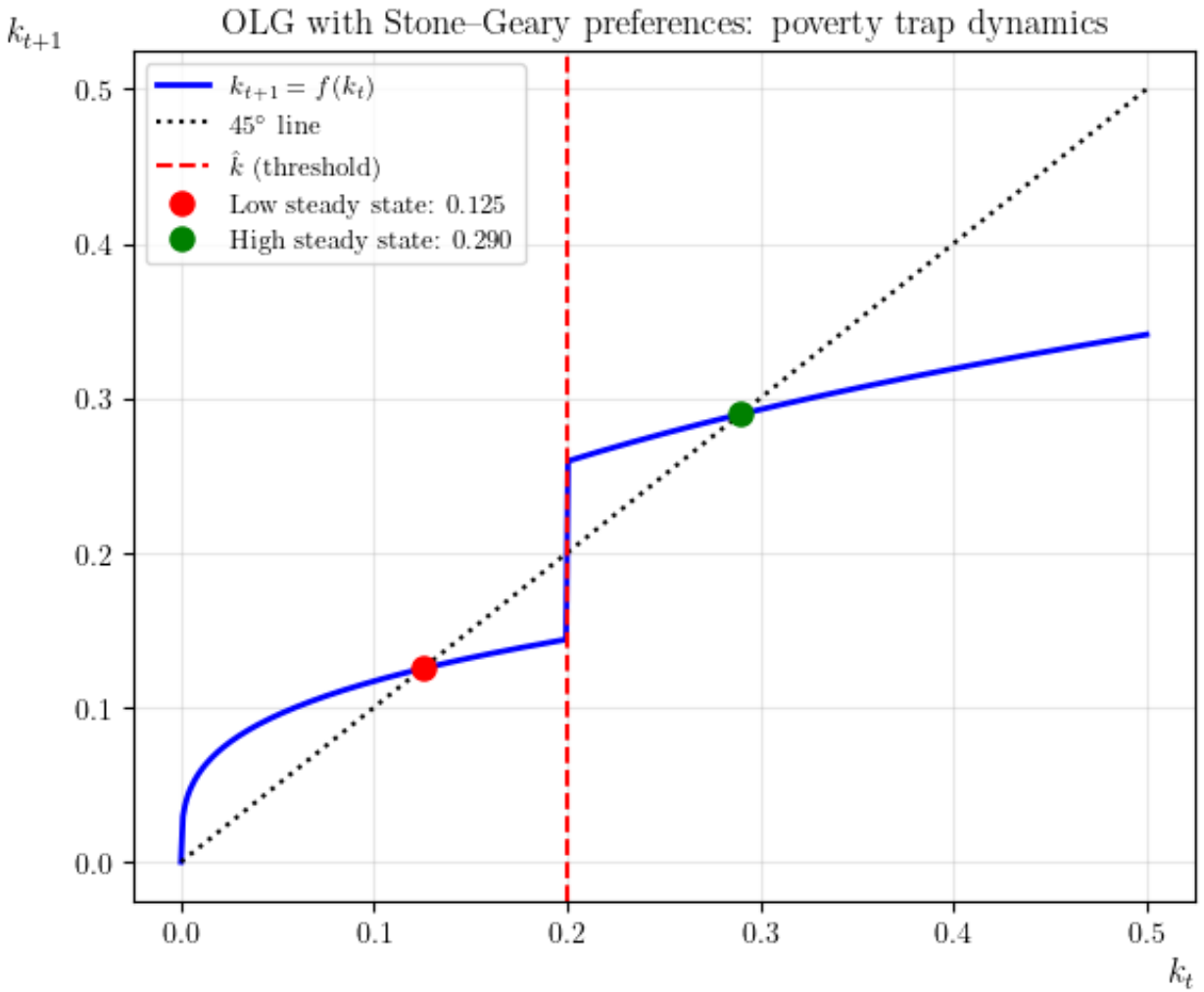
$$k_{t+1} = \begin{cases} \dfrac{s(1-\alpha)}{n} k_t^{\alpha} = A_1 k_t^{\alpha} & k_t < \hat{k} \quad (x_t = 0), \\ \dfrac{s(1-\alpha)}{n - (1-s)\bar{c}/\alpha} k_t^{\alpha} = A_2 k_t^{\alpha} & k_t \geq \hat{k} \quad (x_t = \bar{c}), \end{cases}$$

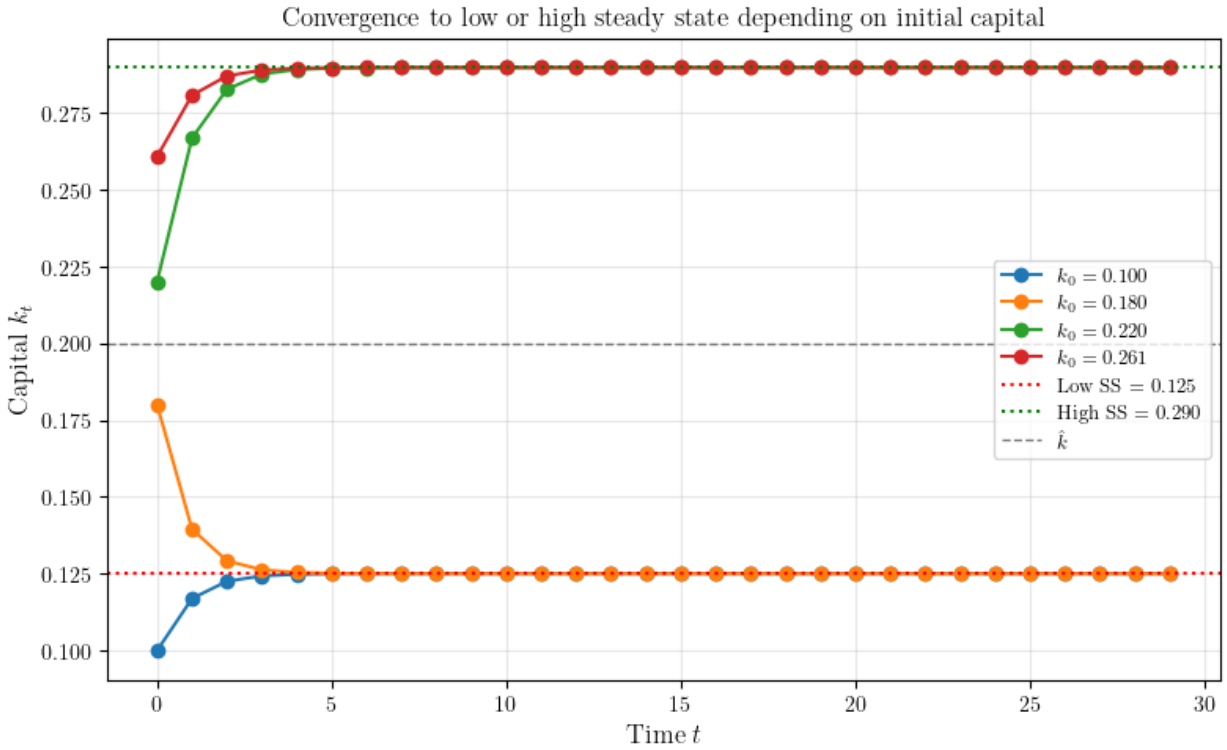
with $A_2 > A_1$.

Two stable steady states when $\hat{k}$ is interior:

$$k_1^* = A_1^{1/(1-\alpha)} < \hat{k} < k_3^* = A_2^{1/(1-\alpha)}.$$

Mechanism: Once $k_t \geq \hat{k}$, households must cover the old-age subsistence requirement $\bar{c} k_{t+1}^{\alpha}$, so they *save more* ($A_2 > A_1$), which sustains the high steady state. Below $\hat{k}$ there is no such requirement, saving is lower and the economy stays at the low steady state. Only a sufficiently large shock can push the economy past $\hat{k}$.


OLG with Stone–Geary preferences: poverty trap dynamics
k_{t+1}
k_t
$k_{t+1} = f(k_t)$
45° line
$\hat{k}$ (threshold)
Low steady state: 0.125
High steady state: 0.290
0.0
0.1
0.2
0.3
0.4
0.5


Convergence to low or high steady state depending on initial capital
Capital k_t
Time t
0.100
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0
5
10
15
20
25
30
$k_0 = 0.100$
$k_0 = 0.180$
$k_0 = 0.220$
$k_0 = 0.261$
Low SS = 0.125
High SS = 0.290
$\hat{k}$

Impatience Trap

A Model of Habit Formation

Utility function (Azariadis, 1996, eq. 14):

$$v(c_1, c_2) = \frac{\log c_1}{\beta(c_1)} + \log c_2 - A(c_1, c_2),$$

where $A(c_1, c_2)$ is chosen to make v continuous at $c_1 = \bar{c}$.

- A state-dependent discount factor: the more you consumed in the past, the more patient you become.
- The discount factor $\beta(c_1)$ is piecewise constant:

$$\beta(c_1) = \begin{cases} \beta_1 & c_1 < \bar{c},\ \beta_1 > 0, \\ \beta_2 & c_1 > \bar{c},\ \beta_2 > \beta_1. \end{cases}$$

When consumption is below $\bar{c}$, the household is impatient (low β). Once consumption crosses $\bar{c}$, the household becomes patient (high β), saving a larger fraction of income.

Optimization

For $c_1 \neq \bar{c}$, the Euler equation is:

$$\frac{1}{\beta_i c_1} = \frac{R_{t+1}}{c_2}, \qquad i \in \{1, 2\}.$$

with budget $c_1 + c_2/R_{t+1} = w_t$. Solving: $c_1 = w_t/(1+\beta_i)$, saving $s_t = s_i w_t$ where $s_i = \beta_i/(1+\beta_i)$.

Kinked saving function:

$$s_t = \begin{cases} s_1 w_t & w_t \leq (1+\beta_1)\bar{c}, \\ w_t - \bar{c} & w_t \in [(1+\beta_1)\bar{c}, (1+\beta_2)\bar{c}], \\ s_2 w_t & w_t \geq (1+\beta_2)\bar{c}. \end{cases}$$

The middle branch reflects the income effect: the household is transitioning from the low-patience to the high-patience regime as it crosses the habit threshold.

5. Habit Formation: Dynamics and Trap

Law of motion (with Cobb–Douglas, $\delta = 1$, $n = 1$):

$$k_{t+1} = \begin{cases} s_1 w(k_t) & k_t \le \bar{k}_1, \\ w(k_t) - \bar{c} & k_t \in [\bar{k}_1, \bar{k}_2], \\ s_2 w(k_t) & k_t \ge \bar{k}_2, \end{cases}$$

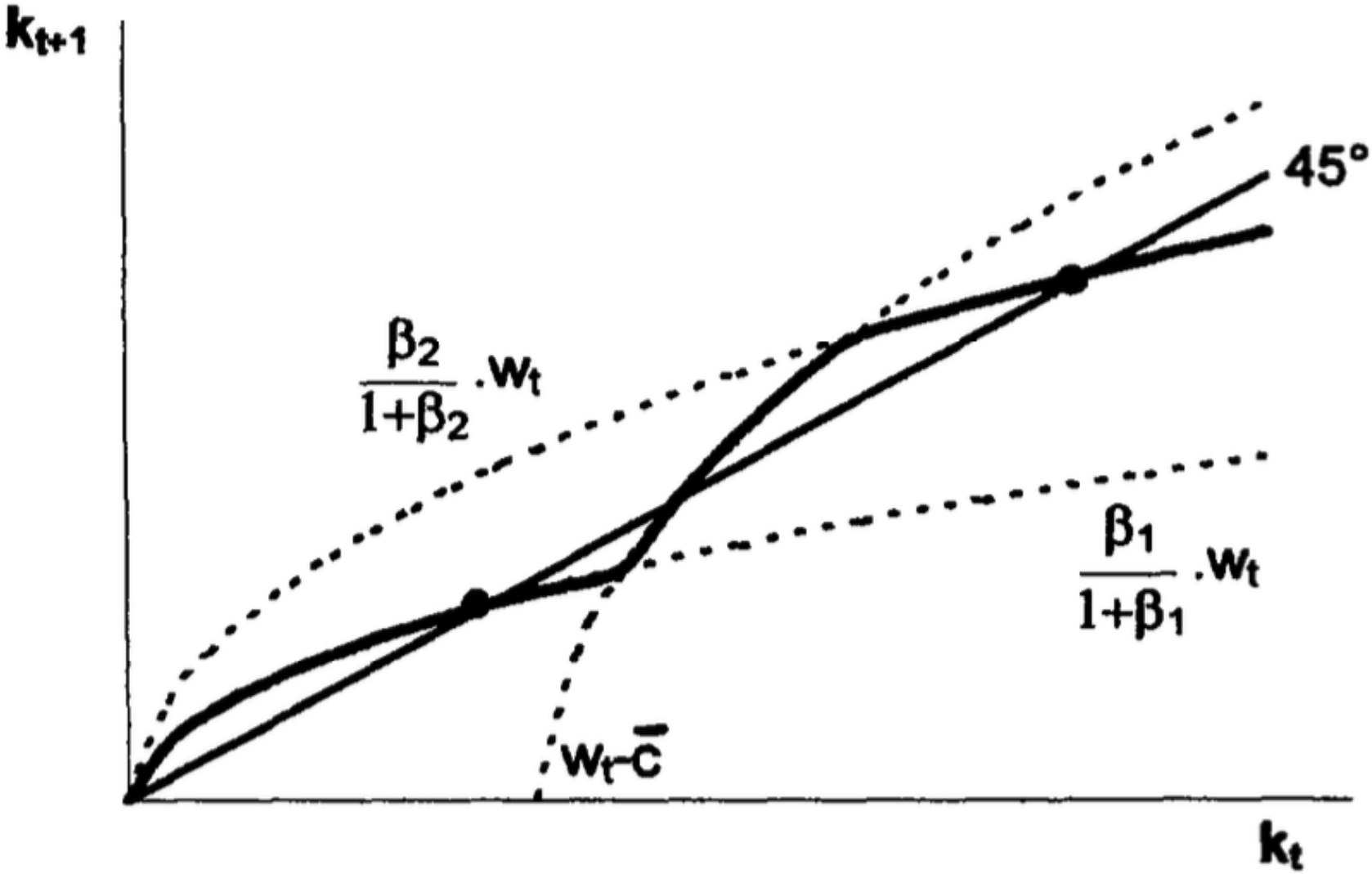
where $w(\bar{k}_i) = (1 + \beta_i)\bar{c}$, $i = 1, 2$.

Two stable steady states: corresponding to saving rates $s_1 < s_2$, separated by an unstable steady state k_U on the middle branch, $w(k_U) - \bar{c} = k_U$.

Impatience trap: The economy converges to the low steady state iff $k_0 < k_U$; in particular whenever

$$w_0 = (1 - \alpha)k_0^{\alpha} < (1 + \beta_1)\bar{c}.$$

A sufficiently large transfer or one-shot productivity shock can release the economy by pushing consumption above the habit threshold $\bar{c}$.


k_{t+1}
45°
$\frac{\beta_2}{1+\beta_2} . w_t$
$\frac{\beta_1}{1+\beta_1} . w_t$
$w_t - \bar{c}$
k_t

External Increasing Returns (Boldrin)

Model Construction

- Overlapping generations with external increasing returns to scale.
- Each producer i uses: $y_t^i = (k_t^{\bar{\gamma}})(k_t^i)^{\alpha}$, where k_t is the economy-wide average capital.
- Private returns: $\alpha < 1$.
- Social returns: $\alpha + \bar{\gamma} > 1$ (increasing).

With symmetry $k_t^i = k_t$, aggregate production is $Y_t = k_t^{\alpha+\bar{\gamma}}$, and

$$w_t = (1-\alpha)k_t^{\alpha+\bar{\gamma}}.$$

Household Optimization

Household problem:

$$\max \ \log c_t + \beta \log d_{t+1}$$

st

$$c_t + s_t = w_t,$$
$$d_{t+1} = R_{t+1} s_t$$

Euler:

$$d_{t+1} = \beta R_{t+1} c_t$$

so

$$c_t = \frac{1}{1+\beta} w_t,$$
$$s_t = \frac{\beta}{1+\beta} w_t,$$

Define $s \equiv \beta/(1+\beta)$; $\mu \equiv \alpha + \bar{\gamma} - 1 > 0$. With constant population and full depreciation, market clearing $k_{t+1} = s_t$ gives the law of motion:

$$k_{t+1} = s(1-\alpha)\, k_t^{1+\mu}.$$

Steady states: solve $k = s(1-\alpha)k^{1+\mu}$.

$$k = 0 \quad \text{(stable)}, \qquad \bar{k} = \left[s(1-\alpha)\right]^{-1/\mu} \quad \text{(unstable)}.$$

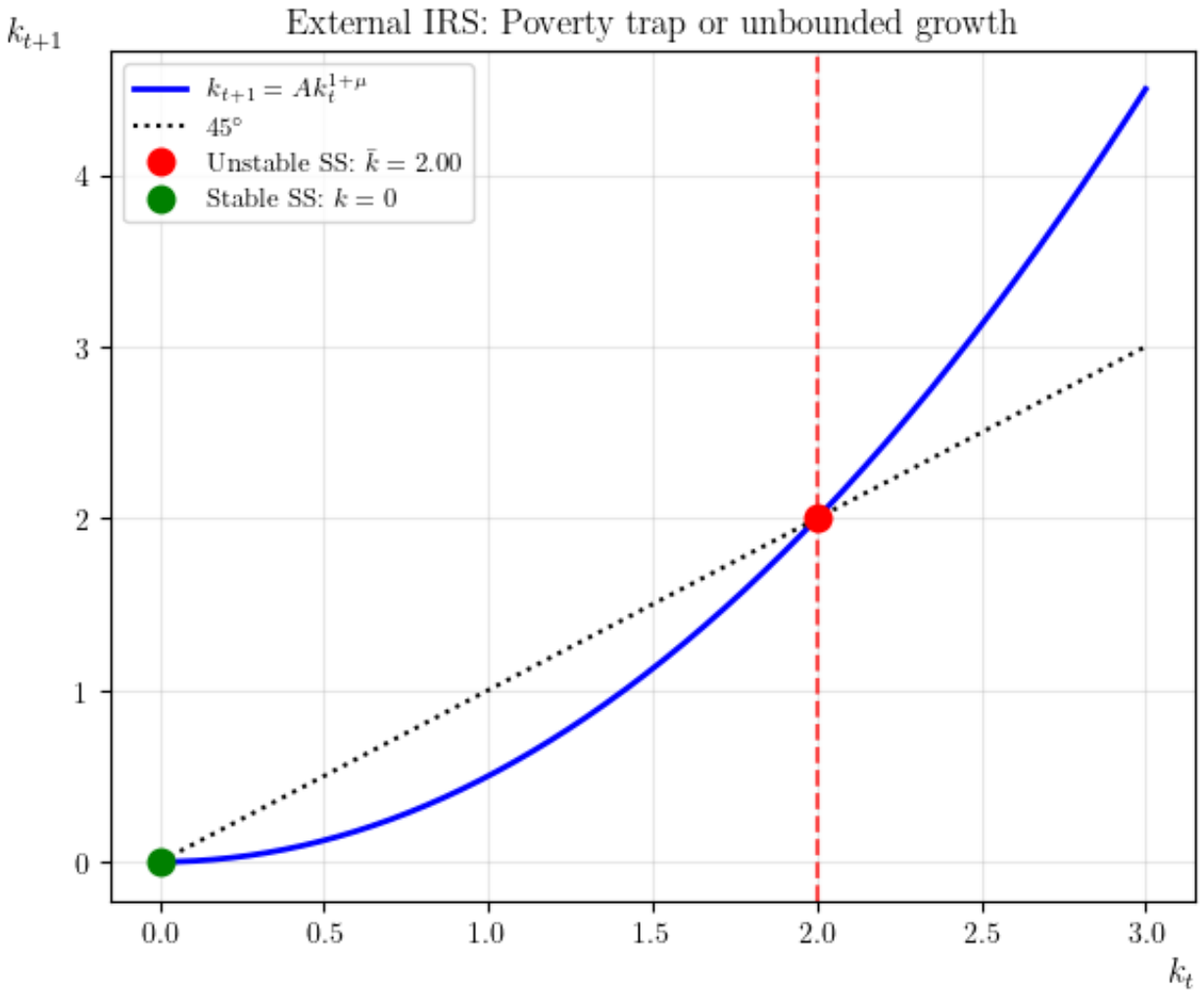
Stability:

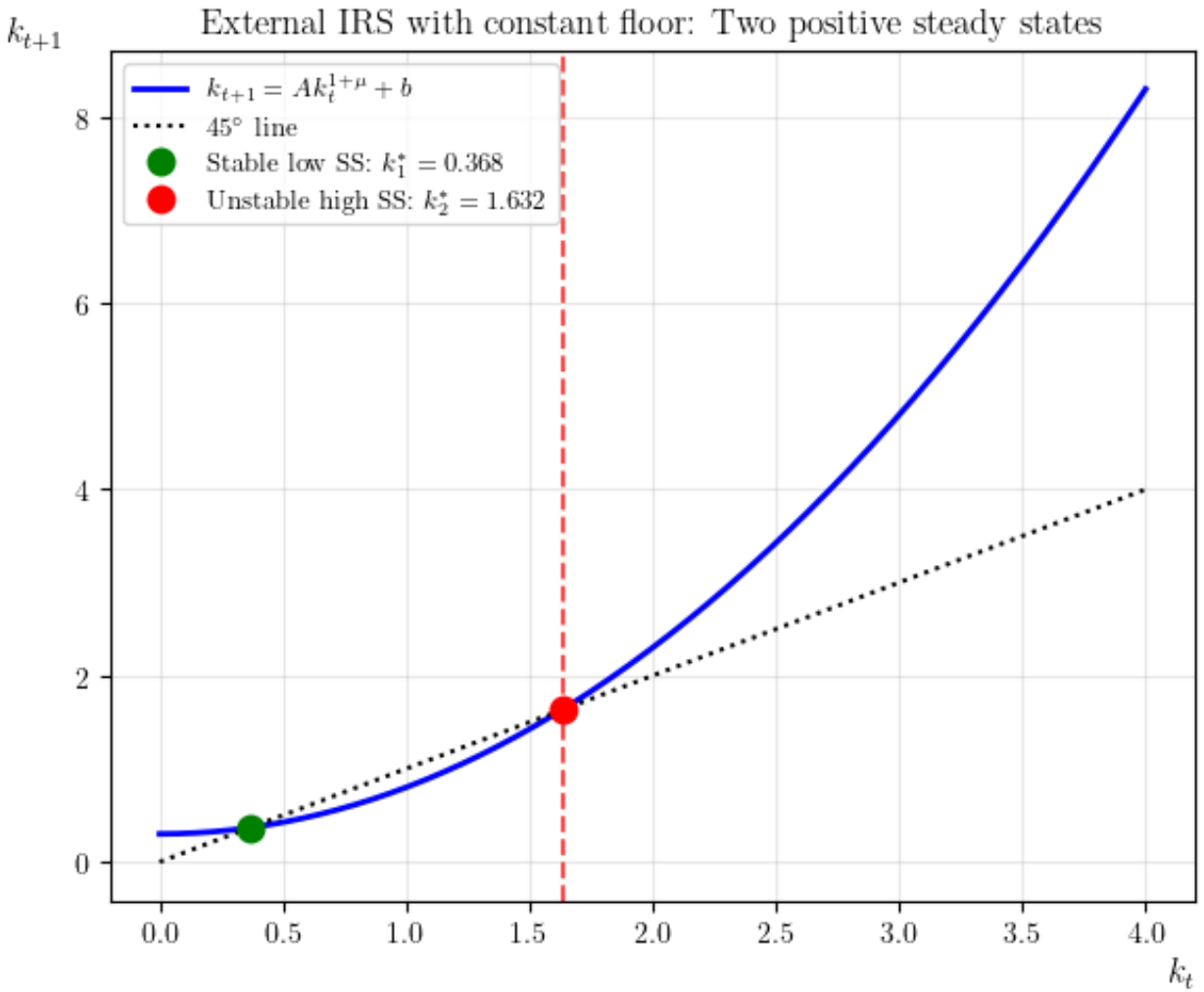
- $k^* = 0$: $\phi'(0) = 0$ (stable sink).
- $\bar{k}$: $\phi'(\bar{k}) = 1 + \mu > 1$ (source).

Dynamics:

- $k_0 \in (0, \bar{k})$: $k_t \to 0 \Rightarrow$ poverty trap.
- $k_0 > \bar{k}$: $k_t \to \infty \Rightarrow$ unbounded growth.

Mechanism: At low capital, the externality is too weak to offset diminishing private returns. Above the threshold $\bar{k}$, the increasing returns dominate and growth becomes self-sustaining. History alone selects the long-run outcome.


External IRS: Poverty trap or unbounded growth
k_{t+1}
$k_{t+1} = Ak_t^{1+\mu}$
45°
Unstable SS: $\bar{k} = 2.00$
Stable SS: $k = 0$
0
1
2
3
4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
k_t


External IRS with constant floor: Two positive steady states
k_{t+1}
$k_{t+1} = Ak_t^{1+\mu} + b$
45° line
Stable low SS: $k_1^* = 0.368$
Unstable high SS: $k_2^* = 1.632$
0
2
4
6
8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
k_t

Biblio

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